Maximally Expressive Constrained Dissipation for 4-Qubit GHZ State Preparation: Mathematical Framework and Experimental Results

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Abstract

We describe an experiment that optimizes a hardware-constrained dissipative channel to steer arbitrary quantum states toward a 4-qubit GHZ target state $|\mathrm{GHZ}_4\rangle = (|0000\rangle + |1111\rangle)/\sqrt{2}$. The constrained channel is composed of local single-qubit and two-qubit CPTP primitives on an all-to-all connectivity graph, including ancilla-reset gadgets. A two-stage optimization (L-BFGS-B on rate parameters, then COBYLA on all 180 parameters) yields a validated fidelity gain of +0.445 from a mean initial fidelity of 0.055 to a final fidelity of 0.499. This is compared against an ideal target-cooling channel that achieves $F(T) = 0.994$. The optimized channel is verified to be completely positive and trace-preserving (CPTP).

1 Problem Setting

We consider the problem of dissipative state preparation: given a target pure state $|\psi_\star\rangle \in \mathbb{C}^d$ with $d = 2^n$, design a dissipative quantum channel $\mathcal{E} : \mathbb{C}^{d\times d} \to \mathbb{C}^{d\times d}$ such that repeated application drives an arbitrary initial density matrix ρ_0 toward the target projector $P_\star = |\psi_\star\rangle\langle\psi_\star|$.

The figure of merit is the *fidelity to the pure target*,

$$F(\rho, \psi_\star) = \langle\psi_\star|\rho|\psi_\star\rangle. \tag{1}$$

In this experiment, $n = 4$ qubits ($d = 16$) and the target is the GHZ state:

$$|\mathrm{GHZ}_4\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle). \tag{2}$$

2 Ideal Teacher: Target-Cooling Channel

As an upper-bound benchmark, we use the *target-cooling channel* defined by the Lindblad jump operators

$$L_k = \sqrt{\gamma}\,|\psi_\star\rangle\langle\psi_k^\perp|, \qquad k = 1, \ldots, d-1, \tag{3}$$

where $\{|\psi_k^\perp\rangle\}_{k=1}^{d-1}$ is any orthonormal basis for the orthogonal complement of $|\psi_\star\rangle$.

2.1 Exact Fidelity Law

The density matrix admits a block decomposition relative to $P = |\psi_\star\rangle\langle\psi_\star|$ and $P_\perp = I - P$:

$$a_0 = \langle\psi_\star|\rho_0|\psi_\star\rangle, \qquad \text{(fidelity, scalar)}, \tag{4}$$

$$B_0 = P_\perp\,\rho_0\,P_\perp, \qquad \text{(orthogonal block)}, \tag{5}$$

$$C_0 = P\,\rho_0\,P_\perp, \qquad \text{(off-diagonal)}, \tag{6}$$

$$D_0 = P_\perp\,\rho_0\,P. \tag{7}$$

Under the Lindblad master equation $\dot{\rho} = \sum_k \gamma\, \mathcal{D}[L_k](\rho)$ (with $H = 0$), the blocks evolve independently:

$$a(t) = 1 - (1 - a_0)\, e^{-\gamma t}, \tag{8}$$

$$B(t) = e^{-\gamma t}\, B_0, \tag{9}$$

$$C(t) = e^{-\gamma t/2}\, C_0, \tag{10}$$

$$D(t) = e^{-\gamma t/2}\, D_0. \tag{11}$$

Equation (8) shows that fidelity converges *exponentially and monotonically* to 1. In the experiment, $\gamma = 1.0$ and $T = 5.0$, giving a theoretical minimum final fidelity of $F(T) \geq 1 - (1 - F_0)\, e^{-5}$.

2.2 Efficient Implementation

This evolution is implemented without forming the full $d^2 \times d^2$ Liouvillian. Each time step uses rank-structured updates with cost $O(d^2)$ per step:

$$\rho(t + \Delta t) = e^{-g}\, \rho(t) + \alpha\, |\psi_\star\rangle\langle\psi_\star| + \beta\, |\psi_\star\rangle\langle w| + \beta\, |v\rangle\langle\psi_\star|, \tag{12}$$

where $g = \gamma\, \Delta t$, $v = \rho|\psi_\star\rangle$, $w = v^*$, and

$$\alpha = (1 - e^{-g}) + 2a_0(e^{-g} - e^{-g/2}), \tag{13}$$

$$\beta = e^{-g/2} - e^{-g}. \tag{14}$$

3 Constrained Dissipative Channel

The constrained model replaces the globally nonlocal teacher jumps (3) with a composition of physically local CPTP primitives. Each time step applies a first-order Trotter product:

$$\mathcal{E}_{\Delta t} \approx \mathcal{E}^{(m)}_{\Delta t} \circ \cdots \circ \mathcal{E}^{(2)}_{\Delta t} \circ \mathcal{E}^{(1)}_{\Delta t}, \tag{15}$$

where each $\mathcal{E}^{(j)}$ is a local Kraus channel acting on one or two qubits. Local channels are applied via tensor contraction on the active qubit indices without forming full $d \times d$ Kraus operators.

3.1 Connectivity

The experiment uses *all-to-all* connectivity on $n = 4$ qubits, yielding $\binom{4}{2} = 6$ edges.

3.2 Channel Primitives

3.2.1 Single-Qubit Channels (per qubit)

Amplitude damping. Kraus operators with parameter $p = 1 - e^{-\gamma_i\, \Delta t}$:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \tag{16}$$

Dephasing. With $e = e^{-2\gamma_i\, \Delta t}$:

$$K_0 = \sqrt{\tfrac{1+e}{2}}\, I, \qquad K_1 = \sqrt{\tfrac{1-e}{2}}\, Z. \tag{17}$$

Depolarizing. With $p = 1 - e^{-\gamma_i\, \Delta t}$:

$$\mathcal{E}(\rho_q) = (1 - p)\, \rho_q + \frac{p}{3}(X\rho_q X + Y\rho_q Y + Z\rho_q Z). \tag{18}$$

3.2.2 Two-Qubit Channels (per edge)

ZZ dephasing. Kraus operators on the 2-qubit subspace:

$$K_0 = \sqrt{\tfrac{1+e}{2}}\, I_4, \qquad K_1 = \sqrt{\tfrac{1-e}{2}}\, (Z \otimes Z), \qquad e = e^{-2\gamma_{ij}\,\Delta t}. \tag{19}$$

Bell-state pumping. With $p = 1 - e^{-\gamma_{ij}\,\Delta t}$ and target Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$:

$$\mathcal{E}(\rho_{ij}) = (1-p)\,\rho_{ij} + p\,|\Phi^+\rangle\langle\Phi^+|. \tag{20}$$

Parity pumping. Projects onto the even-parity sector of the ZZ eigenbasis:

$$\mathcal{E}(\rho_{ij}) = (1-p)\,\rho_{ij} + p\,\Pi_{\text{even}}\,\rho_{ij}\,\Pi_{\text{even}} + p\,\Pi_{\text{odd}}\,\rho_{ij}\,\Pi_{\text{odd}}, \tag{21}$$

where $\Pi_{\text{even/odd}}$ project onto the ± 1 eigenspaces of $Z \otimes Z$.

3.2.3 Ancilla-Reset Gadgets

1-qubit ancilla reset (15 parameters per qubit). A 4×4 unitary on (system qubit, ancilla) is generated as

$$U = \exp\Big(-i \sum_{k=1}^{15} \theta_k\, G_k\Big), \tag{22}$$

where $\{G_k\}_{k=1}^{15}$ is the traceless 2-qubit Pauli basis $\{I, X, Y, Z\}^{\otimes 2} \setminus \{I \otimes I\}$. The ancilla starts in $|0\rangle$ and is traced out, giving two Kraus operators:

$$K_m = \langle m|_{\text{anc}}\, U\, |0\rangle_{\text{anc}}, \quad m \in \{0,1\}, \qquad K_m \in \mathbb{C}^{2\times 2}. \tag{23}$$

2-qubit ancilla reset (15 parameters per edge). An 8×8 unitary on (qubit i, qubit j, ancilla) is generated from the 3-qubit Pauli basis (15 of 63 generators), yielding two 4×4 Kraus operators per edge.

3.3 Parameter Summary

Table 1 shows the parameter layout.

Table 1: Parameter layout for the maximally expressive 4-qubit configuration.

Channel family	Count	Type	Total
Amplitude damping	4	rate	4
Dephasing	4	rate	4
Depolarizing	4	rate	4
ZZ dephasing	6	rate	6
Bell pumping	6	rate	6
Parity pumping	6	rate	6
1Q ancilla reset	4×15	angle	60
2Q ancilla reset	6×15	angle	90
Total			**180**

All rate parameters $\gamma_i \in [0, \gamma_{\max}]$ with $\gamma_{\max} = 2.0$. Ancilla-reset angle parameters θ_k are clipped to $[-2, 2]$ during optimization.

4 Optimization

4.1 Sigmoid Reparameterization

Rate parameters are mapped from unconstrained reals via a sigmoid:

$$\gamma = \frac{\gamma_{\max}}{1 + e^{-x}}, \qquad x = \log \frac{\gamma}{\gamma_{\max} - \gamma}. \tag{24}$$

4.2 Loss Function

The dissipative trajectory is simulated for $S = 20$ steps over $t \in [0, T]$ with $T = 5.0$. The loss evaluated on a training set of $|\mathcal{S}|$ initial states $\{\rho_0^{(s)}\}$ is:

$$\mathcal{L} = \frac{1}{|\mathcal{S}|} \sum_{s=1}^{|\mathcal{S}|} \sum_{j=1}^{S} w_j \left(1 - F^{(s)}(t_j)\right) + \lambda_r \sum_i \Big(\frac{\gamma_i}{\gamma_{\max}}\Big)^2 + \lambda_\theta \sum_k \theta_k^2, \tag{25}$$

where $F^{(s)}(t_j) = \langle \psi_\star | \rho^{(s)}(t_j) | \psi_\star \rangle$ is the fidelity at time step j for training state s.

Curve weights. Exponentially decaying weights emphasize early dynamics:

$$w_j = \frac{\tilde{w}_j}{\sum_k \tilde{w}_k}, \qquad \tilde{w}_j = e^{-\alpha\, t_j}, \qquad \alpha = \frac{2}{T}. \tag{26}$$

Regularization. $\lambda_r = 0.005$ penalizes large rates; $\lambda_\theta = 0.01$ (Stage 2 only) penalizes large ancilla angles.

4.3 Two-Stage Optimization

The 180-parameter landscape is navigated with a staged approach:

Stage 1: Rate-only (L-BFGS-B). All 150 angle parameters are fixed to zero. Only the 30 rate parameters are optimized using L-BFGS-B with automatic differentiation through finite differences. This stage finds a good basin in the rate subspace.

Stage 2: Full optimization (COBYLA). All 180 parameters are jointly optimized using the derivative-free COBYLA method, warm-started from Stage 1. The initial angles are set to small random values $\theta_k \sim \text{Uniform}(-0.3, 0.3)$ to break symmetry.

4.4 Training States

Three initial states form the training set:

(i) two random mixed density matrices $\rho_{\text{rand}}^{(s)}$ generated as $\rho = AA^\dagger / \operatorname{Tr}(AA^\dagger)$ with $A \in \mathbb{C}^{16\times 16}$ having i.i.d. complex Gaussian entries,

(ii) the maximally mixed state $\rho = I/16$.

Their initial fidelities to $|\text{GHZ}_4\rangle$ are $F_0 \in \{0.063, 0.075, 0.063\}$.

5 CPTP Validation

The optimized channel is verified to be a valid quantum channel through two tests on the full $d \times d = 16 \times 16$ system:

Trace preservation. For all basis operators $|i\rangle\langle j|$:

$$\max_{i,j} |\mathrm{Tr}(\mathcal{E}(|i\rangle\langle j|)) - \delta_{ij}| \leq \epsilon_{\mathrm{TP}}. \tag{27}$$

Complete positivity. The Choi matrix

$$C_{\mathcal{E}} = \sum_{m,n=0}^{d-1} \mathcal{E}(|m\rangle\langle n|) \otimes |m\rangle\langle n| \in \mathbb{C}^{d^2 \times d^2} \tag{28}$$

is Hermitianized and its minimum eigenvalue is checked: $\lambda_{\min}(C_{\mathcal{E}}) \geq -\epsilon_{\mathrm{CP}}$.

6 Experimental Results

6.1 Optimization Trajectory

- **Stage 1** (L-BFGS-B, rates only): 1736 function evaluations over 628 s. Loss decreased from 0.853 to 0.585. Mean training fidelity at $t = T$: $F(T) = 0.500$.
- **Stage 2** (COBYLA, all 180 parameters): 2000 function evaluations over 834 s. Loss decreased from initial ~ 0.997 (disrupted by angle randomization) to 0.599. Mean training fidelity at $t = T$: $F(T) = 0.499$.

Total wall time: 1462 s.

6.2 Validation Results

Four held-out states (three random mixed, one random pure) were used for validation. Results are summarized in Table 2.

Table 2: Experimental results: constrained channel vs. ideal teacher.

Metric	Training	Validation
Mean initial fidelity F_0	0.067	0.055
Mean final fidelity $F(T)$	0.499	0.499
Mean fidelity gain ΔF	+0.432	+0.445
Gain range	[+0.424, +0.437]	[+0.423, +0.453]
Max trace deviation	1.4×10^{-14}	1.3×10^{-14}
Teacher $F(T)$ ($\gamma = 1$)	0.994	
Constrained / Teacher ratio	0.503	
Trace-preserving	True ($\epsilon = 1.7 \times 10^{-15}$)	
Completely positive	True ($\lambda_{\min} = 6.4 \times 10^{-12}$)	

6.3 Key Observations

1. **Substantial fidelity gain.** The constrained channel increases fidelity from ~ 0.06 to ~ 0.50, a gain of +0.44—well above the initial value and demonstrating that local dissipative primitives can meaningfully steer states toward a globally entangled target.
2. **Fidelity ceiling at $F \approx 0.5$.** Despite 180 tunable parameters and all-to-all connectivity, the optimized channel saturates near $F = 0.5$. For the GHZ state $(|0000\rangle + |1111\rangle)/\sqrt{2}$,

this corresponds to the population being concentrated in the correct computational-basis support $\{|0000\rangle, |1111\rangle\}$ but without the correct relative phase (coherence). Local channels can pump population but struggle to establish the long-range coherence $\langle 0000|\rho|1111\rangle = 1/2$ required for unit fidelity.

3. **Generalization.** Training and validation fidelities match closely (0.499 vs. 0.499), indicating the optimized channel generalizes across initial states rather than overfitting to the training distribution.

4. **Gap to the teacher.** The constrained channel achieves roughly 50% of the teacher's fidelity (0.503 ratio). This quantifies the cost of restricting from globally nonlocal jump operators (Eq. 3) to local Kraus primitives.

5. **CPTP integrity.** The Trotterized composition of local channels preserves trace to machine precision (10^{-15}) and complete positivity ($\lambda_{\min} > 0$), confirming that the optimized map is a valid quantum channel.

6. **Rate-only optimization captures most of the gain.** Stage 1 already achieves $F(T) = 0.500$ using only 30 rate parameters. Stage 2 with all 180 parameters does not significantly improve this, suggesting the fidelity ceiling is a fundamental limitation of the channel family rather than an optimization artifact.

7 Discussion

The central finding is a *phase-coherence barrier*: local dissipative channels composed via Trotterization can redistribute population among computational basis states but cannot efficiently generate the long-range quantum coherence characteristic of GHZ-type entanglement.

Mathematically, the GHZ fidelity decomposes as

$$F = \langle \mathrm{GHZ}|\rho|\mathrm{GHZ}\rangle = \frac{1}{2}(\rho_{0000,0000} + \rho_{1111,1111}) + \mathrm{Re}(\rho_{0000,1111}). \tag{29}$$

The population terms $\rho_{0000,0000} + \rho_{1111,1111}$ can be driven toward 1 by local amplitude-damping and parity-pumping channels. However, the coherence term $\mathrm{Re}(\rho_{0000,1111})$ requires correlated phase alignment across all 4 qubits simultaneously—a task for which the local channel primitives in the model have limited expressive power.

The saturation at $F \approx 0.5$ is consistent with the optimized channel achieving full population transfer ($\rho_{0000,0000} + \rho_{1111,1111} \to 1$) while the coherence term remains near zero ($\mathrm{Re}(\rho_{0000,1111}) \approx 0$).

This result suggests several directions for future work:

- *Coherence-generating primitives:* Adding explicit multi-qubit unitary rotations or engineered cross-qubit phase channels.

- *Measurement-feedback protocols:* Using mid-circuit measurements to condition subsequent dissipative steps.

- *Product-state targets:* Testing whether the fidelity ceiling disappears for targets without long-range entanglement.